Mini Data Science Project

GitHub Repository Link: [https://github.com/SafnasGitHub/mini-data-science-project.git]

Title: Student Performance Prediction Using Machine Learning

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1.Introduction

In education, data science and machine learning are increasingly significant, utilizing vast amounts of student data such as study time, attendance, and grades to analyze factors affecting academic performance and to predict future outcomes. Educators can identify students needing extra support and make data-informed decisions. A project employed the Student Performance Dataset to apply data science techniques, involving data preparation, exploration, visualization, and model evaluation to derive meaningful insights and forecast final grades.

1.1 Problem Formulation

Student academic performance is impacted by prior academic performance, attendance, and study habits. This project explores how these variables relate to final grades, specifically examining the predictability of study time, absences, and prior grades on final outcomes.

The objectives that are discussed in this project are as follows:

1. To examine the relationship between study habits and absences and the performance of students.
2. To define which variables have the strongest correlation with final grades.
3. To forecast final grades of students using machine learning models.

Research Question

What is the relationship between study habits and academic factors and student performance and can machine learning models be precise predictors of student final grades?

1.2 Data Acquisition

The project utilizes the Student Performance Dataset from the UCI machine learning repository, focusing on the student-mat.csv file, which contains demographic, social, and academic characteristics of secondary school students specifically related to their performance in mathematics.

Dataset Source : UCI Machine learning repository - Student performance dataset.

Dataset Description : The information contained in the dataset includes: School, Gender, Age, Address type, Family size, Parent education, Study time, Absences, First period grade (G1), Second period grade (G2), Final grade (G3).

Dataset Size : Number of rows is 395 and number of columns is 33. G3 is the target variable in this project as it is an end of course grade of every student.

2.Data Analysis

The dataset was cleaned and explored before implementing the machine learning methods, and it was necessary to learn the structure of this dataset and find the interesting patterns.

2.1 Data Cleaning and Preprocessing

- Loaded data using Pandas.
- Used sep=';' to read the CSV file.
- Checked rows, columns, and features.
- Examined missing values, duplicates, and data types.
- Found the data suitable for analysis.
- Selected study time, absences, G1, G2, and G3 for prediction.

2.2 Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand student performance distribution and the relationship between key variables, resulting in five visualizations to address the research question

Figure 1: Distribution of Final Grades

The histogram of end grades shows that most students fall within the middle range, indicating a moderate distribution in performance, with fewer students receiving very low or very high marks.

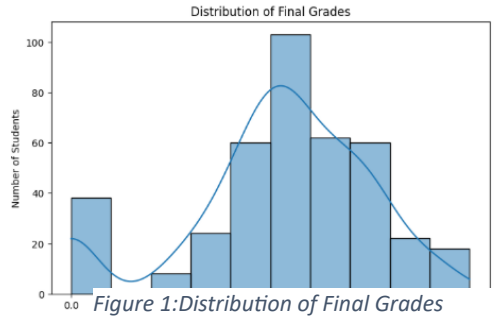

Figure 1:Distribution of Final Grades

Figure 2: Final Grade vs Time of Study

The scatter plot indicates a positive relationship between study time and final grades, showing that students who study more tend to achieve higher grades. While the relationship is not linear, study habits are significant factors in academic achievement according to the graph.

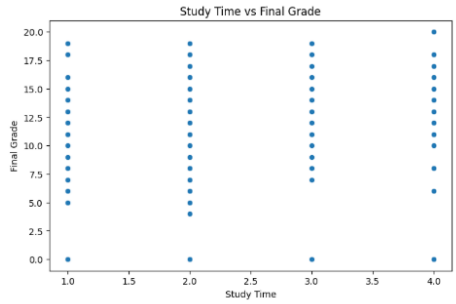

Figure 2:Final Grade vs Time of Study

Figure 3: absenteeism and final grade

The comparison graph shows that increased student absences correlate with lower final grades, indicating that regular attendance significantly impacts academic performance and generally leads to better outcomes.

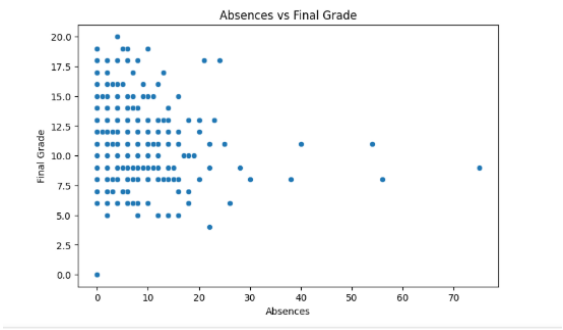

Figure 3:absenteeism and final grades

Figure 4: Heatmap Correlation

The correlation heatmap revealed that the final grade (G3) is positively correlated with previous grades (G1 and G2), indicating that earlier academic performance effectively predicts final outcomes. Additionally, study time showed a significant, albeit weaker, relationship with the final grade.

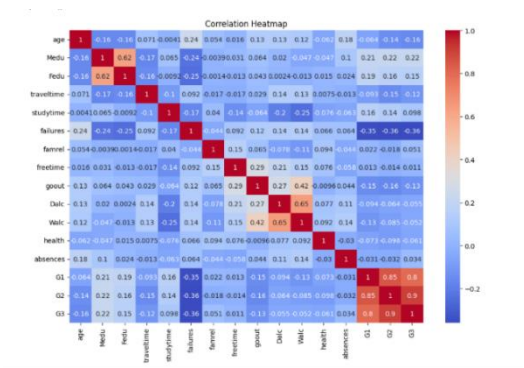

Figure 4:Heatmap Correlation of Numerical

Figure 5: Study Time Category Final Grade

The box plot used to compare the categories of study time and the final grades aid in generalizing the study results in terms of performance variations among various study habits. Higher categories of students in study time tended to perform better in median performance than the lower categories of students. It is corroborated by the hypothesis that higher levels of study effort are added value to academic performance.

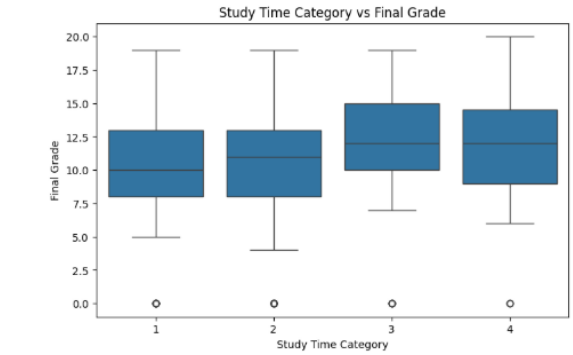

Figure 5:: Study Time Category Final Grade

Overall, the exploratory analysis revealed a moderate correlation between parental affection and child-parent attachment, as well as between child-parent attachment and identified psychopathological symptoms.

3.ML / AI Methods

Once the exploratory was done, machine learning models were used to predict the final grades of the students. The target variable G3 is a numerical variable and hence regression-related methods were chosen.

Predictor Variables : These were the features in the form of the following variables: Study time, absences, G1, G2

Target Variable : G3 (Final Grade)

The data was divided into 70 percent training and 30 percent testing data with the random state of 42. The models were built on the training data and the performance of prediction assessed on the testing data.

The following three regression models were applied in this project:

3.1 Linear Regression

The baseline model for predicting student grades was linear regression, achieving an R^2 score of 0.81, indicating it explains 81% of the variation in final grades. The model has a mean absolute error of 1.30 and a mean squared error of 4.25, demonstrating its effectiveness as a predictor.

```
X = data[['studytime', 'absences', 'G1', 'G2']]
y = data['G3']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
```

Figure 6:Linear Regression Model

3.2 Random Forest Regressor

An ensemble learning model, the Random Forest Regressor, demonstrated superior performance in predicting student final grades with an R2 Score of 0.87, MAE of 1.05, and MSE of 2.90, by integrating multiple decision trees to capture complex data relationships.

```
X = data[['studytime', 'absences', 'G1', 'G2']]
y = data['G3']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

rf = RandomForestRegressor(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
```

Figure 7:Random Forest Regressor

3.3 Decision Tree Regressor

An ensemble learning model, the Random Forest Regressor, demonstrated superior performance in predicting student final grades with an R2 Score of 0.87, MAE of 1.05, and MSE of 2.90, by integrating multiple decision trees to capture complex data relationships.

```
X = data[['studytime', 'absences', 'G1', 'G2']]
y = data['G3']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

dt = DecisionTreeRegressor(random_state=42)
dt.fit(X_train, y_train)

y_pred_dt = dt.predict(X_test)
```

Figure 8: Decision Tree Regressor

3.4 Model Performance Comparison

The following table summarizes the performance of all three models:

Model	R^2 Score	MAE	MSE
Linear Regression	0.81	1.30	4.25
Random Forest Regressor	0.87	1.05	2.90
Decision Tree Regressor	0.83	1.16	3.85

The comparison clearly shows that Random Forest Regressor was the best-performing model.

4.Results and Interpretation

The project shows that machine learning and data science can effectively analyze and predict student performance, with previous grades, absences, and study time significantly affecting final grades. Random Forest emerged as the most accurate model, yielding the highest accuracy and minimal error. Overall, the project illustrates the potential of machine learning in understanding student performance and enhancing academic decision-making.

5.Reflection

This project provided me with experience in the entire process of data science, including data collection, cleaning, visualization and machine learning. The difficulty of loading the dataset correctly in Python and selecting the most useful features to make predictions was one of them. The other problem was how to make proper graphs and be able to explain them. This assignment has taught me that information cleaning is extremely crucial prior to implementing machine learning models. I also got to know that visualizations can be used to obtain patterns and aid in a better analysis. Altogether, the project improved my understanding of Python and showed how data science can provide valuable insights from real educational data.

7.Reference

[1] F. Chollet, *Deep Learning with Python*. Shelter Island, NY, USA: Manning Publications, 2018.

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